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Lecture 3
THEORY OF PRODUCTION AND COST

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THEORY OF PRODUCTION

Introduction

- One of the fundamental questions that society faces is
  - How to produce goods and services that individuals consume

- Generally, much of the output consumed by individuals is produced by firms
  - This is the case because firms have certain advantages over households with respect to large-scale production

Introduction

- In our analysis of supply, we have assumed that the supply curve is upward slopping
  - this implies that higher prices encourage firms to supply more

- But how much will firms choose to supply at each price?

Introduction

- Firms in deciding *how much to produce/supply*
  - do not only combine resources so as to minimize total production costs, but most importantly they compare expected TR against total cost

- Based on these they select the output level that maximizes total output

Short-run Versus Long Run

- In the short run it is assumed that the supply of some inputs is variable while others are fixed.

- The long run is a time period long enough for all inputs to be varied.

The Production Function

- A *production function* describes the various combinations of factor inputs needed to produce a particular output.

- The factor inputs are measured in physical rather than in monetary units.

- Mathematically production function can be given as; Q=f(K,L).

The Production Function

- K=fixed factor (capital)
- L=variable factor (labour)
- In the short-run where capital input is fixed, the production function becomes;

$$Q = f(\bar{K}, L)$$

The Production Function

- If more and more of the variable factor is applied on the fixed factor,
  - the productivity of it would decline with time as additions to output from the extra labour will begin to diminish.

- At this stage we say that *the law of diminishing marginal returns* has set in.

The Short-Run Production Function: Total, Average and Marginal Products

- *Total product* (*TP*) is the maximum output that a given quantity of labour can produce.

- *Average product* of labour (AP_L) is equal to total output divided by the quantity of labour employed (i.e. $AP_L = Q/L$).

The Short-Run Production Function: Total, Average and Marginal Products

- *Marginal product* (MP_L) of labour
  - is the increase in total output that results from a one-unit increase in the quantity of labour employed *with all other inputs remaining the same* (i.e. $MP_L = \Delta Q / \Delta L$).

- *Average product of capital* (AP_K) is total output divided by the quantity of fixed input or capital (K)

The Short-Run Production Function: Total, Average and Marginal Products

- Marginal product of capital (MP_K) is the change in total output (ΔQ) divided by a change in capital (ΔK) (i.e. $MP_K = \Delta Q / \Delta K$).

- These can be illustrated graphically in the next diagram;

The Short-Run Production Function: Total, Average and Marginal Products

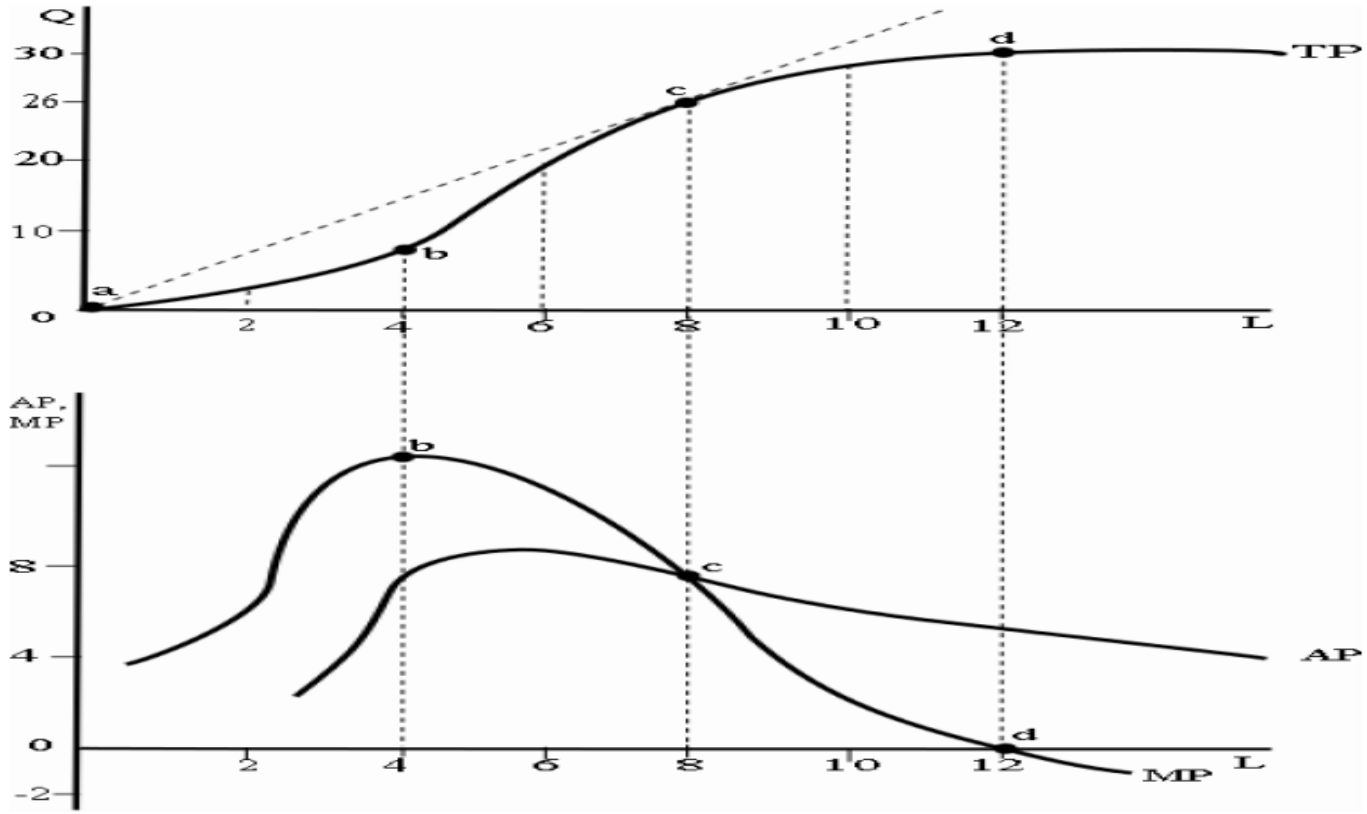


Figure 4.1: Total Product, Average Product and Marginal Product

The Long-Run Analysis of Production

- Firms make several decisions including techniques and scale of production in the long run.

- If a firm decides to double all its inputs
  - output would just double (*constant returns to scale*)
  - more than double (*increasing returns to scale*)
  - less than double (*decreasing returns)*

Returns to Scale and Homogeneity of the Production Function

- Suppose a firm's initial inputs and output are given by
- $Y_0=f(K,L)$
- If all factors are increased by the same proportion λ, we obtain
- $Y_t=f(\lambda K, \lambda L)$
- If λ can be factored out then we can express Y_t as a function of λ

Returns to Scale and Homogeneity of the Production Function

- $Y_t = \lambda^v f(K,L)$

- If λ can be factored out, then the production function is referred to as a *homogeneous function*

- However, if λ cannot be factored out, the production function is *non-homogeneous*.

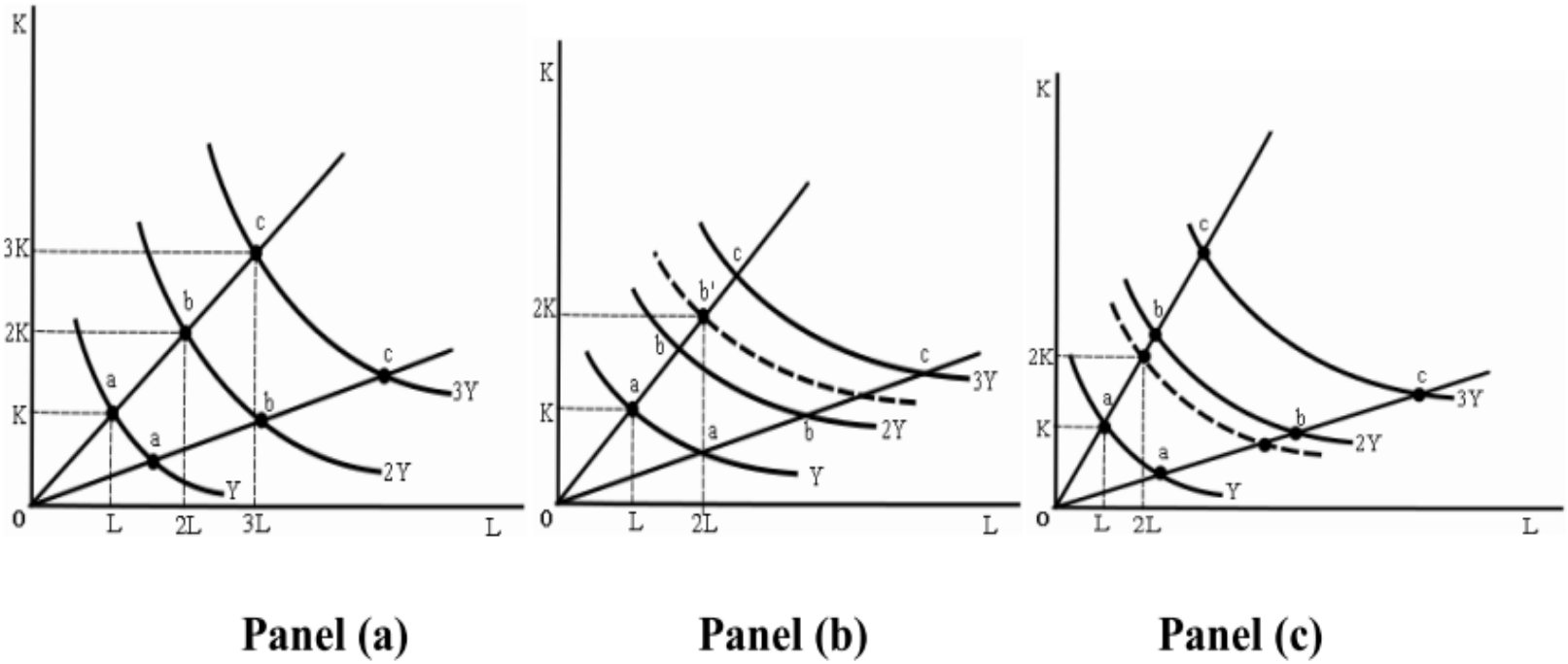
Returns to Scale and Homogeneity of the Production Function

- If v=1, the production function is *linear homogeneous* and exhibit *constant returns to scale*
- If v<1, then we have *decreasing returns to scale*
- If v>1, then we have *increasing returns to scale*
- Illustration with Examples

Returns to Scale of Homogeneous Production Function: A Graphical Approach

- For homogeneous production functions
  - returns to scale can be represented graphically by the distance between successive isoquants.
  - For *constant returns to scale* the distance between successive isoquants is constant.
  - For *increasing returns to scale,* the distance between successive isoquants increases.
  - For *decreasing returns to scale*, the distance between successive isoquants decreases.
- These can be illustrated in the next Figure;

Returns to Scale of Homogeneous Production Function: A Graphical Approach



Panel (a) **Panel (b)** **Panel (c)**

Figure 4.2: Constant, Increasing and Decreasing Returns to Scale

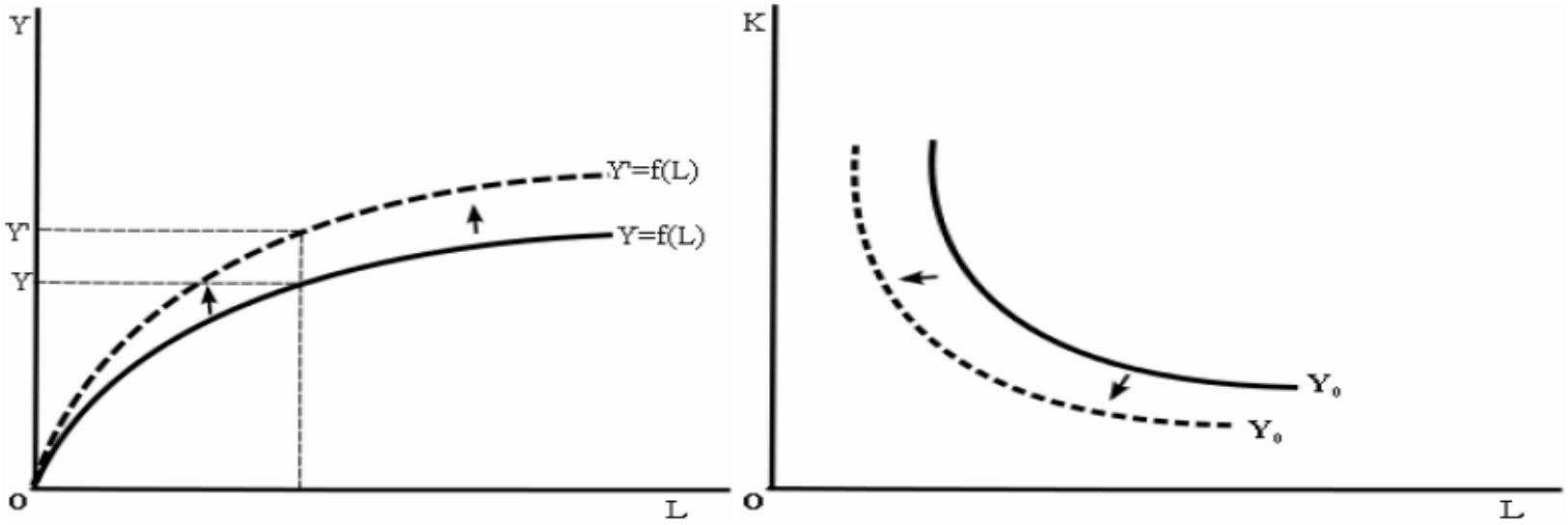
Technical Progress and the Production Function

- Techniques of production change
  - when new knowledge or invention and more efficient methods of production become available.

- Effect of technological change
  - can be shown with an upward shift of the production function, or a downward movement of the production *isoquant*.

- An *isoquant* is a line joining combinations of factors of production (K,L) that generate the same level of output

Technical Progress and the Production Function



Panel (a) **Panel (b)**

Figure 4.3: Technical Progress and the Production Function

Technical Progress and the Production Function

- Three types of technical progress, depending on its effect on the rate of substitution of the factors of production are known.

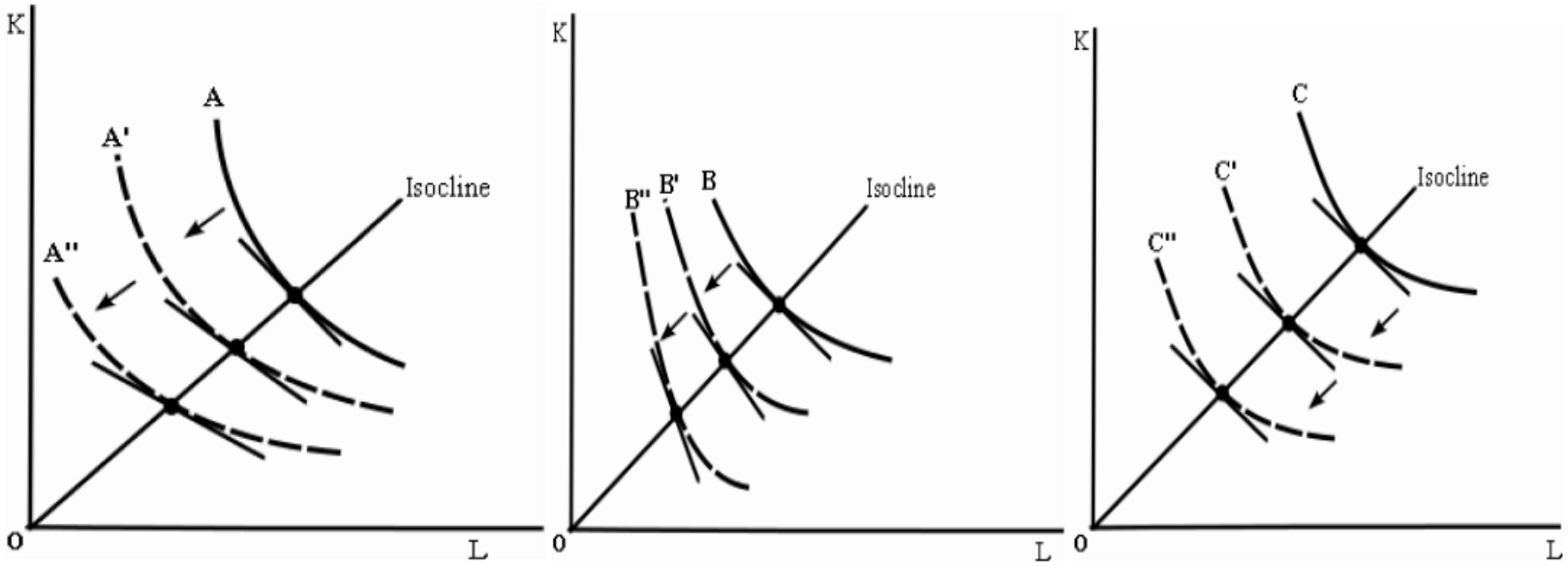
- *Capital-deepening technical progress* occurs if, along the line which passes through the origin on which the K/L ratio is constant, (MP_K) increases more than the (MP_L).

Technical Progress and the Production Function

- *Labour-deepening technical progress* occurs if,
  - along the line which passes through the origin on which the K/L ratio is constant, MP_L increases more than the MP_K.

- *Neutral technical progress* occurs if,
  - along the line which passes through the origin on which the K/L ratio is constant, the MP_L and MP_K increase by the same amount.
- If firms can achieve the least-cost combination of inputs, then they are *efficient in production*.

Technical Progress and the Production Function



Panel (a) Panel (b) Panel (c)

Figure 4.4: Capital-deepening, Labour-deepening, and Neutral Technical Progress

COSTS OF PRODUCTION

Introduction

- We have shown earlier
  - how firms can vary factors of production in the short-run and long run in order to increase output
- We also know that firms incur costs when they buy factor inputs used in producing output.
- Thus firms in deciding how much to produce
  - need to know the precise level of costs in order to maximize profit
- It becomes imperative, therefore that
  - firms understand what constitute costs in both the *short-run* and the *long run*

Economic and Accounting Costs

- The term *cost is* used differently by economists and accountants.

- Economists often use the concept of *opportunity cost* when measuring costs.

- Economic costs consists of two aspects:
- Implicit costs
- Explicit costs.

Economic and Accounting Costs

- *Implicit costs*
  - represent the value of alternatives forgone in the production of a good, irrespective of whether money payments are made.

- *Explicit costs*
  - involve the money payment for using productive resources. It is also known as *accounting cost.*

The Short-Run Analysis of Costs

- In the short run,
  - firms produce with both *fixed* and *variable inputs*
  - this implies they incur both fixed and variable costs

- *Fixed costs* do not vary with increases or decreases in the level of firms output.

- *Variable costs* increase or decrease with changes in the level of firms' output.

The Law of Increasing Cost

- The law of increasing cost states that
  - beyond some level of output or once all factors of production are at maximum output and efficiency in the short run,
  - variable costs (and therefore total cost) would increase more on average than output.

- This results because of diminishing marginal return to the variable input

The Short-Run Cost Curves: Total, Average and Marginal Costs

- A firm's short-run *total cost* (*TC*) *is* the cost of all its factors of production.

- It is made up of the sum of *total fixed* (*TFC*) and *variable costs* (*TVC*). *TC* = *TFC* + *TVC*

- These costs concepts are illustrated graphically in the next diagram;

The Short-Run Cost Curves: Total, Average and Marginal Costs

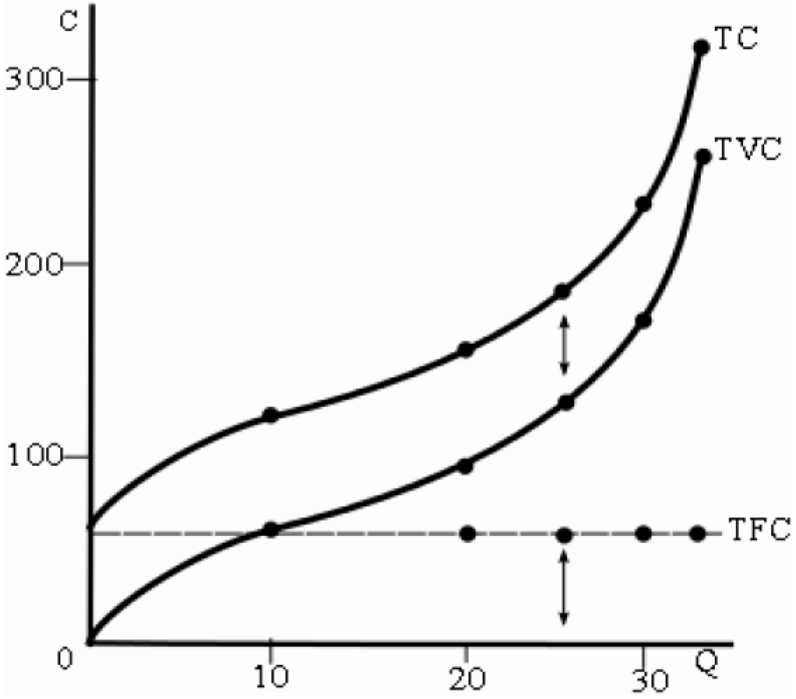


Figure 5.1: Costs Curves: Total Costs, Total Fixed Costs, and Total Variable Costs

The Short-Run Cost Curves: Total, Average and Marginal Costs

- Both *TC* and *TVC* curves slope upward because total variable cost increases as output increases.

- The *TVC* curve starts from the origin because with zero output, no variable factor is used.

- The shape of *TVC* (and hence *TC*) is influenced by the *law of diminishing returns*.

The Short-Run Cost Curves: Total, Average and Marginal Costs

- *TVC* initially increases at a decreasing rate and then begins to increase at an increasing rate as output increases.

- Average fixed cost (*AFC*) is total fixed cost per unit of output (*TFC*/*Q*).

- Average variable cost (AVC) is total variable cost per unit of output (*TVC*/*Q*).

- Average cost (*AC*) is total cost per unit of output (*TC*/*Q*).

The Short-Run Cost Curves: Total, Average and Marginal Costs

- Marginal cost ($MC=\Delta TC/\Delta Q$)
  - is the extra cost incurred in producing one more unit of the output

- All *MC*s are variable, since when output increase no additional *FC* is incurred.

- *AFC*, *AVC*, *AC* and *MC* is illustrated graphically next

The Short-Run Cost Curves: Total, Average and Marginal Costs

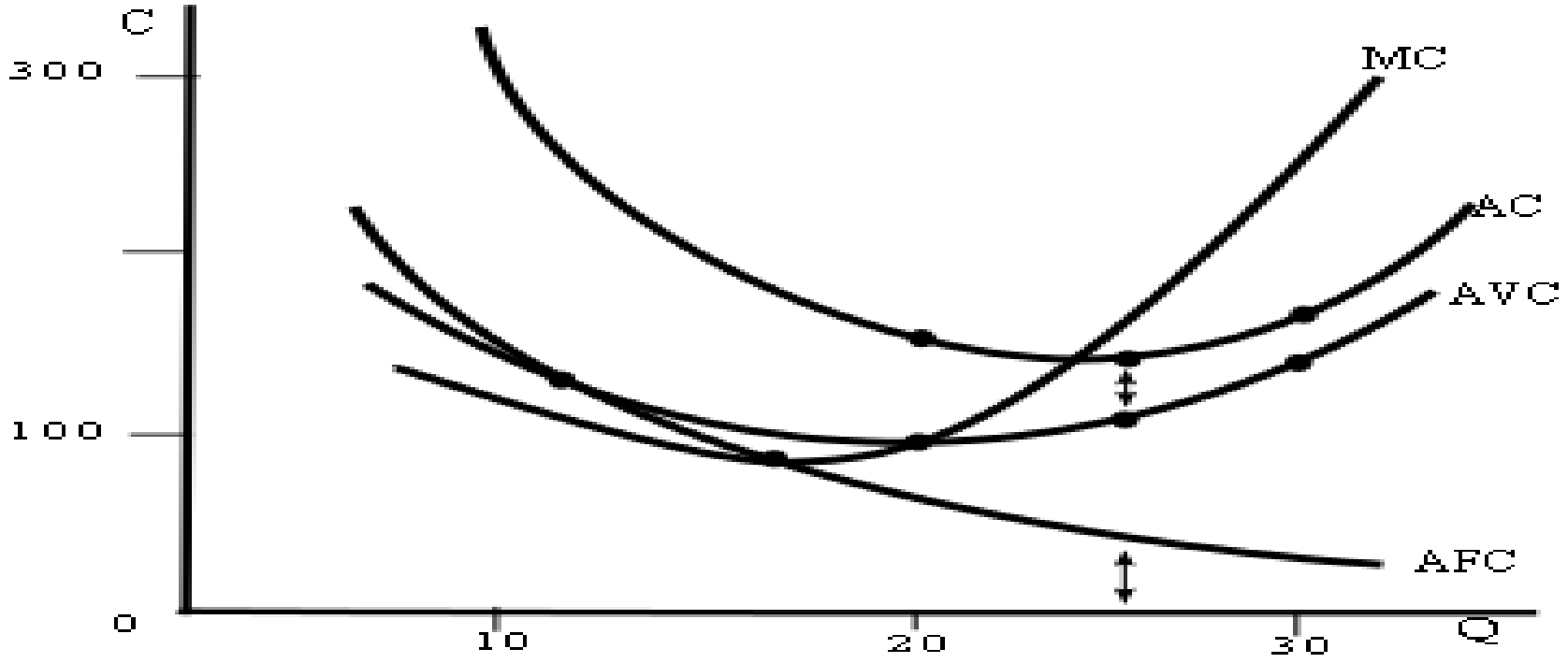


Figure 5.2: Average and Marginal Cost Curves

The Short-Run Cost Curves: Total, Average and Marginal Costs

- Both *TC* and *TVC* curves slope upward
  - because total variable cost increases as output increases.

- The *TVC* curve starts from the origin because with zero output, no variable factor is used

- The shape of *TVC* (and hence *TC*) is influenced by the *law of diminishing returns*.

The Short-Run Cost Curves: Total, Average and Marginal Costs

- *TVC* initially increases at a decreasing rate and then begins to increase at an increasing rate as output increasing

- The *AFC* curve slopes downward, because as output increases, the same constant *TFC* is spread over a larger and larger output.

- Both the *AC* and the *AVC* curves are U-shaped.

The Short-Run Cost Curves: Total, Average and Marginal Costs

- As output increases, both *AFC* and *AVC* decreases (thus *AC* also decreases).
  - the *AC* curve slopes downward initially

- However, as output increases further, and diminishing returns set in, *AVC* begins to increase more quickly, than the decrease in *AFC*, thus *AVC* reaches its minimum earlier than *AFC*

The Short-Run Cost Curves: Total, Average and Marginal Costs

- The *MC* curve is also U-shaped.
- *MC* initially decreases at low levels of output
  - due to economies of scale through greater specialisation,
  - but increases later due to the *law of diminishing returns*
- *AC* and *MC* are relatad
  - because as additions to output cost less than the average, *AC* must decline
- This implies that if *MC* is less than *AC*, then *AC* must be falling, and vice versa

Product and Cost Curves

- The *U*-shaped behaviour of the *AC* and *MC* curves are closely related to the hump-shaped behaviour of the *AP* and *MP* curves.

- The next diagram highlights the links between technology (product curves) and cost (cost curves).

Product and Cost Curves

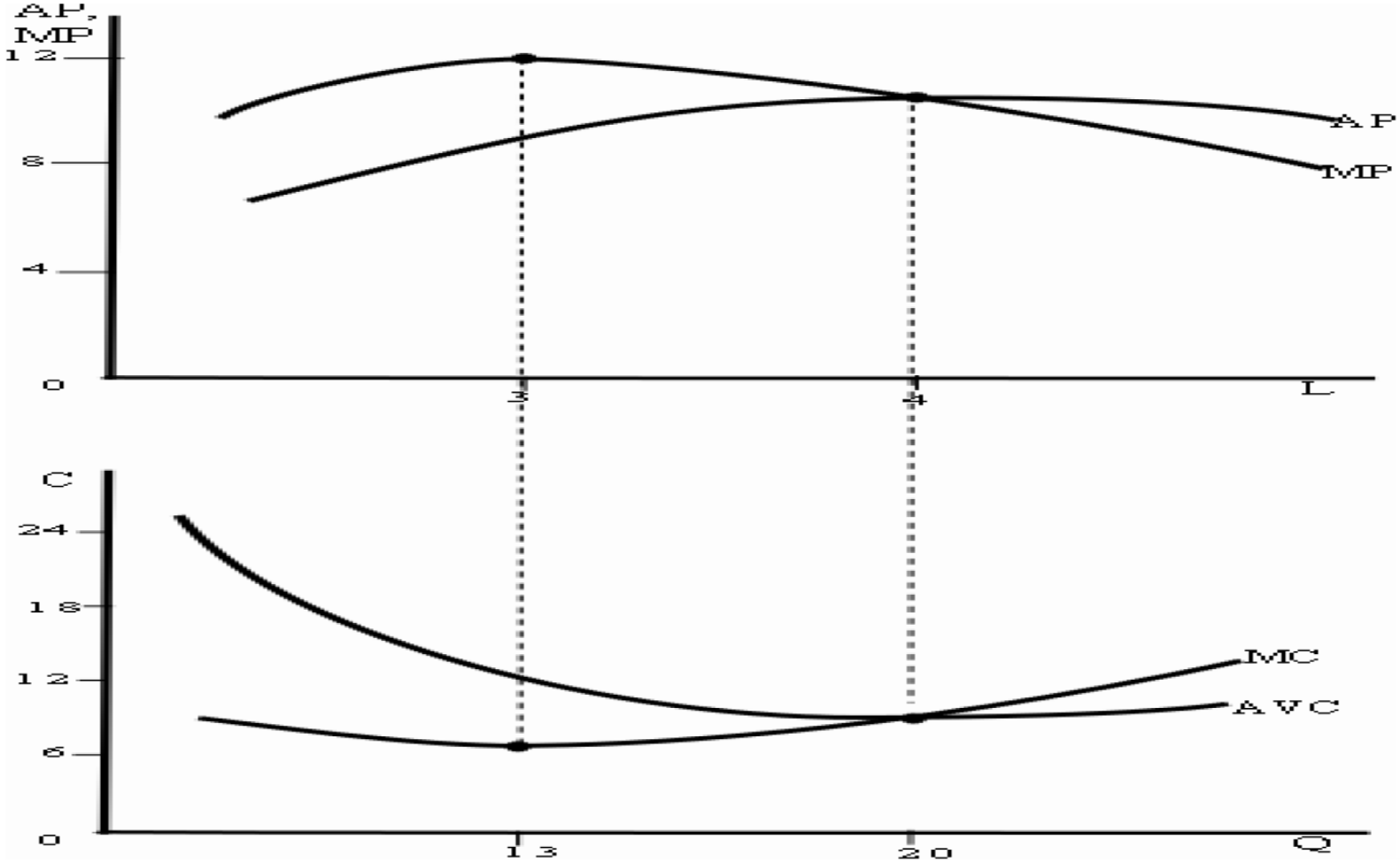


Figure 5.3: Product and Cost Curves

Product and Cost Curves

- Technology and prices of factors of production can determine
  - the position of a firm's short run cost curves

- A change in technology that increases productivity will shift
  - the *TP*, *AP* and *MP* upwards and the *TC*, *AC* and *MC* downwards as technology will lower the cost of production

The Long Run Analysis of Costs

- The long run is considered to be the period of planning for firms
  - because they have no fixed factors and hence no fixed cost.

- The production function enables firms to determine the minimum long run total cost (*LTC*) of producing various levels of output.

The Long Run Analysis of Costs

- The firm's long run average cost (*LRAC*) and long run marginal cost (*LMC*) can then be derived from its *LTC*.

- *LRAC* curves normally slope *downward* at first and then upward (they are U-shaped).

- This can be illustrated in the next diagram

The Long Run Analysis of Costs

Economies & Diseconomies of Scale

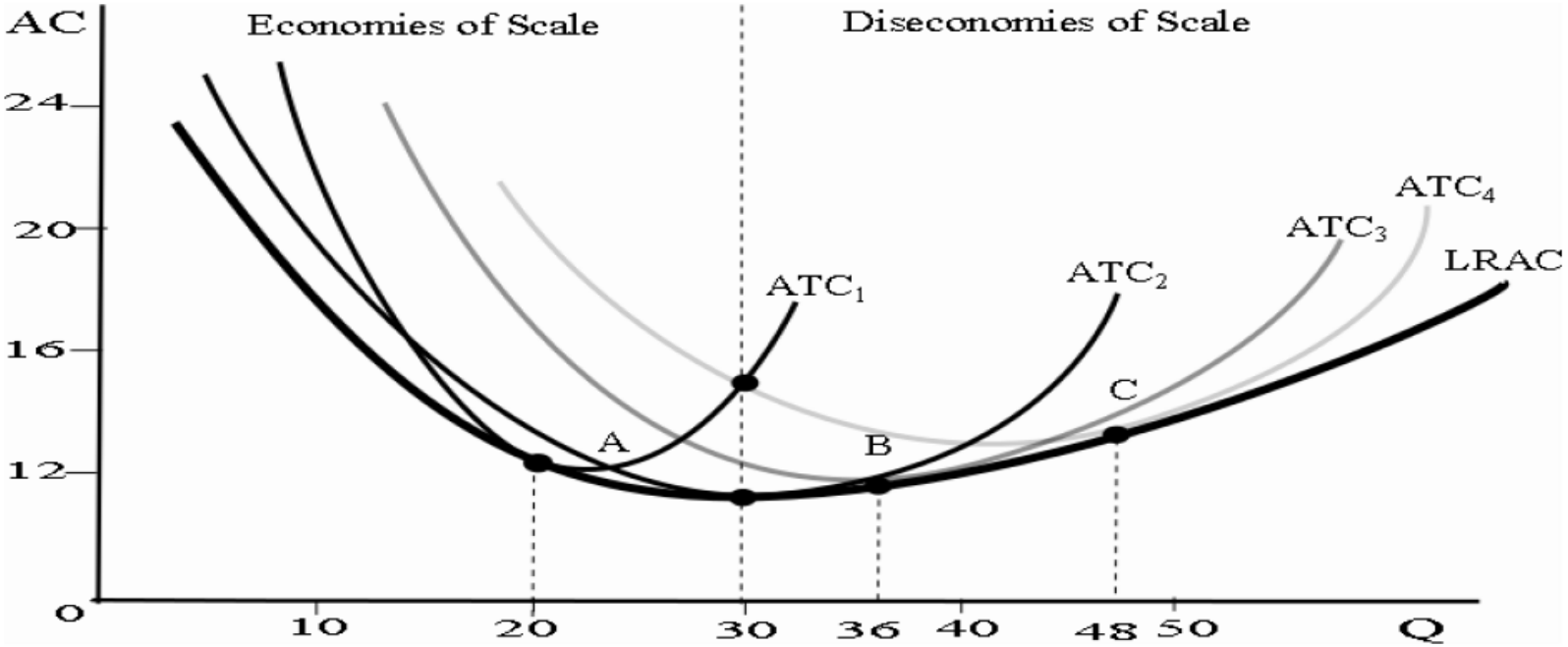


Figure 5.4 The long run and short run average cost curves

Economies and Diseconomies of Scale

- *Economies of scale* are the cost advantages that a firm enjoys as it expands its scale of operations in the long run.

- When economies of scale are present, the *LRAC* curve slopes downward.

- Economies of scale are distinguished into *pecuniary and real economies* of scale.

Economies and Diseconomies of Scale

- *Pecuniary economies*
  - benefits enjoyed by firms as a result of paying lower prices for production inputs.

- *Real economies*
  - those advantages associated with a reduction in physical quantity of inputs, raw materials, various types of capital and labour.

The End